

Decision tree learnig

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier, export_text

# Read the dataset
data = pd.read_excel("/content/Student_Attendance_Dataset.xlsx")

# Display the dataset
print("==== DATASET =====")
print(data)

# Separate features and target
X = data.iloc[:, :-1].copy()
y = data.iloc[:, -1].copy()

# Encode categorical columns
encoders = {}

for column in X.columns:
    if X[column].dtype == 'object':
        le = LabelEncoder()
        X[column] = le.fit_transform(X[column])
        encoders[column] = le

# Encode target column
target_encoder = LabelEncoder()
y = target_encoder.fit_transform(y)

# Create Decision Tree Classifier
model = DecisionTreeClassifier(
    criterion='entropy',
    random_state=42
)

# Train the model
model.fit(X, y)

# Print Decision Tree Rules
print("\n==== DECISION TREE RULES =====")
```

```
print(export_text(model, feature_names=list(X.columns)))
```

```
# Training Accuracy
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accuracy = model.score(X, y)
```

```
print("\nTraining Accuracy:", accuracy)
```

```
# Prediction using first record
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```
sample = X.iloc[[0]]
```

```
prediction = model.predict(sample)
```

```
print("\nPrediction for First Record:",  
      target_encoder.inverse_transform(prediction)[0])
```

OUTPUT

```
*** ===== DATASET =====
```

	StudentID	StudentName	Department	Attendance	Assignment	InternalMarks	\
0	S001	Aarav	IT	Yes	Yes	82	
1	S002	Diya	CSE	No	Yes	68	
2	S003	Rahul	ECE	Yes	No	75	
3	S004	Sneha	IT	Yes	Yes	91	
4	S005	Kiran	MECH	No	No	55	
5	S006	Priya	CSE	Yes	Yes	79	
6	S007	Vikram	EEE	No	Yes	61	
7	S008	Anu	IT	Yes	Yes	88	
8	S009	Rohit	CIVIL	No	No	58	
9	S010	Meena	ECE	Yes	Yes	84	

	LabCompleted	Eligible
0	Yes	Yes
1	No	No
2	Yes	Yes
3	Yes	Yes
4	No	No
5	Yes	Yes
6	No	No
7	Yes	Yes
8	No	No
9	Yes	Yes